

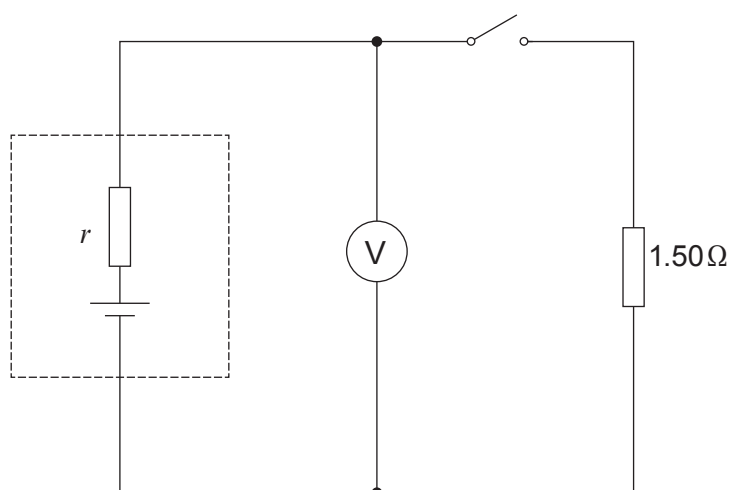
5. (a) Define the *potential difference* between two points in an electric circuit. [2]

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- (b) A cell of emf 1.62 V and internal resistance, r , is included in the circuit shown.



- (i) State the expected reading on the voltmeter when the switch is open. [1]
- (ii) With the switch closed the voltmeter reads 1.38 V . Show in clear steps that the cell's internal resistance, r , is approximately $0.3\ \Omega$. [3]

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- (iii) 750 J of the cell's energy is dissipated **in total** while the switch is closed. Calculate the time for which the switch is closed. [2]

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- (iv) Calculate the voltmeter reading when the 1.50Ω resistor is replaced by a 0.75Ω resistor, and the switch is closed. [2]

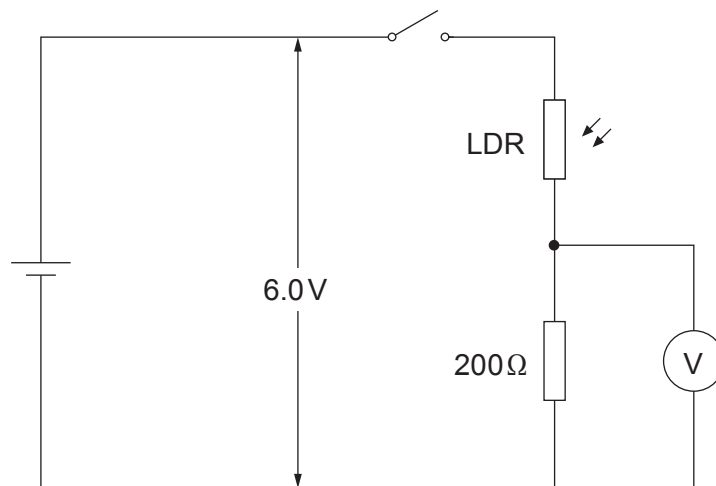
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- (c) The circuit shown includes a light-dependent resistor (LDR), whose resistance decreases as the intensity of light falling on it increases. A 6.0V supply of negligible internal resistance is used.



- (i) Calculate the voltmeter reading when the resistance of the LDR is 850Ω and the switch is closed. [2]

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- (ii) Explain, in clear steps, whether the voltmeter reading will increase or decrease when the intensity of light is increased. [2]

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